

Meeting summary — Jakarta EE Future Directions, July 10, 2025

Gavin King joined the interest group to discuss forward-looking persistence work for Jakarta EE, especially Jakarta Data, Jakarta Persistence, and Jakarta Query. He reviewed Jakarta Data 1.0 as a new stateless repository model with stronger compile-time validation through annotation processing, including type-safe validation of repository methods and JPQL/HQL queries.

Gavin also summarized Jakarta Persistence 3.2 improvements, emphasizing many smaller enhancements that collectively improve type safety, programmatic startup/configuration, schema export, validation, and testing workflows. He then outlined planned Jakarta Data 1.1 work, including a restriction/constraint API, projections, CDI-based event notifications, asynchronous/reactive repositories, and early support for stateful repositories.

For Jakarta Persistence 4.0, Gavin highlighted the proposed “entity agent,” a stateless persistence API similar to Hibernate’s stateless session, intended to better align Jakarta Persistence with Jakarta Data’s stateless repository model. He also introduced early work on Jakarta Query, which aims to define a common query language with clearer compliance levels and more rigorous semantics.

The discussion closed with a broader future-facing topic: Java’s role in AI, quantum computing, and scientific computing. Gavin argued that Java needs stronger linear algebra foundations, comparable to Python’s NumPy ecosystem, to remain relevant in emerging technical areas. Neil suggested inviting Gavin back for a future quantum-focused discussion, and Michael offered to connect him with Barry Bird, who has been working on quantum computing.